

# Full-Database Heterogeneity Validation

Sphere Region Method, Pflugfelder HI & WEPL ( $n = 69\,295$ ,  $r = 15\text{ mm}$ )

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# Recap & New Approach

**Last week** ( $n = 3\,950$ ): Spherical neighbourhood around BP  $\rightarrow$  six advanced HU metrics + 3-D Sobel edge metrics, correlated against GPR. Best predictor:  $S_{95, \text{BP}}$  (Spearman  $\rho = -0.391$ ).

## **This week: three major developments**

- 1 **Full-database scale-up**: run on the complete ADoTA training set ( $n = 69\,295$  beamlets,  $18\times$  larger).
- 2 **Sphere region method**: replace the variable-length BP-range extraction with a *fixed* sphere of radius  $r$  centred on the Bragg-peak voxel. Two radii tested:  $r = 15$  mm (complete) and  $r = 5$  mm (in progress).
- 3 **New metrics**: Pflugfelder heterogeneity index (HI) and Water-Equivalent Path Length (WEPL) mean & std.

Dataset:  $n = 69\,295$  beamlets from the ADoTA training set, beam energies 70 MeV to 250 MeV. Model: DoTA\_v3\_grid\_search\_v11.

# Methodology Change: Sphere Region

## Previous (BP-range method):

- Proximal boundary  $z_{\min}$ : walk backward from BP to where  $\text{IDD} < 0.50 \text{IDD}_{\max}$ .
- Distal boundary  $z_{\max}$ : walk forward to  $\text{IDD} < 0.10 \text{IDD}_{\max}$ .
- Region spans all slices in  $[z_{\min}, z_{\max}]$ .
- *Variable length*: shallow beams  $\approx 10$  mm, deep beams  $\approx 50$  mm.

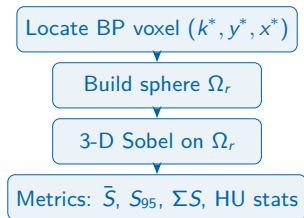
## New (sphere method):

- Locate BP voxel  $(k^*, y^*, x^*)$  from  $\arg \max_k \text{IDD}[k]$  and lateral dose peak.
- Include all voxels within distance  $r$  of BP:

$$\Omega_r = \{(k, y, x) \mid \|\mathbf{p} - \mathbf{p}^*\|_{\text{mm}} \leq r\}$$

where  $\mathbf{p} = (k\Delta z, y\Delta y, x\Delta x)$ .

- *Fixed geometry*: radius  $r$  is a single hyperparameter; independent of beam energy.



|           | BP-range                  | Sphere             |
|-----------|---------------------------|--------------------|
| Geometry  | variable                  | fixed              |
| Length    | $\propto E$               | $r = \text{const}$ |
| Hyperpar. | $\alpha, \beta$ fractions | $r$ [mm]           |
| Lateral   | full beam                 | within $r$         |

# New Metrics: Pflugfelder HI & WEPL

## Water-Equivalent Path Length (WEPL)

For each lateral ray  $(y, x)$  through the CT:

$$\text{WEPL}(y, x) = \Delta z \sum_{k=0}^{k^*} \text{RSP}(k, y, x)$$

RSP = relative stopping power (HU  $\rightarrow$  RSP via Schneider calibration);  $k^*$  = BP slice index.

WEPL map  $\in \mathbb{R}^{H \times W}$ : water-equivalent range to BP per lateral ray.

### Output scalars per beamlet:

- `wep1_mean` =  $\mu(\text{WEPL})$  [mm]
- `wep1_std` =  $\sigma(\text{WEPL})$  [mm]
- `pflugfelder_hi` = HI

## Heterogeneity Index (HI) [1]:

$$\text{HI} = \frac{\sigma(\text{WEPL})}{\mu(\text{WEPL})}$$

Computed over the flux footprint (10 % threshold).

### Interpretation:

- $\text{HI} \approx 0$ : homogeneous path (pure water).
- $\text{HI} \gg 0$ : large lateral stopping-power variation  $\rightarrow$  range uncertainty.

**Motivation:** operates in dose space (mm of water) rather than image space (HU) — complementary to Sobel and HU metrics.

Pflugfelder, Wilkens, Oelfke, *Phys. Med. Biol.* 53(6):1689, 2007.

# Dataset & Model Inference ( $n = 69\,295$ )

## Dataset:

- $n = 69\,295$  beamlets (full ADoTA training set)
- 853 skipped (zero flux or duplicate)
- Energy range: 70 MeV to 250 MeV
- Grid:  $160 \times 30 \times 30$  voxels,  $2 \times 2 \times 2$  mm

**Model:** ADoTA v3 (DoTA\_v3\_grid\_search\_v11)

Gamma: 2%/2 mm, 10 % lower dose cutoff

## GPR distribution (overall):

Mean = 98.83%,  $\sigma = 1.47\%$

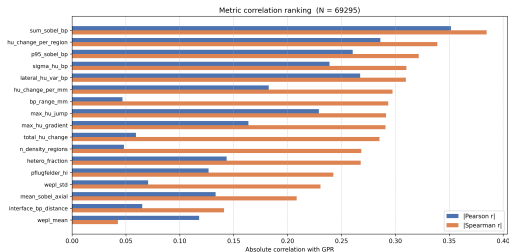
Range: [57.22%, 100.00%]

## Energy bin distribution:

| Bin (MeV) | $n$    | GPR (%)          |
|-----------|--------|------------------|
| 70–100    | 16 163 | $98.46 \pm 1.85$ |
| 100–130   | 15 869 | $98.76 \pm 1.58$ |
| 130–160   | 16 995 | $98.91 \pm 1.41$ |
| 160–190   | 13 881 | $99.21 \pm 0.96$ |
| 190–220   | 3 231  | $99.16 \pm 0.63$ |
| 220–250   | 3 156  | $98.63 \pm 0.76$ |

Pattern from last week confirmed at  $18\times$  scale: lower energies show larger GPR variance.

# Overall Correlation Ranking ( $n = 69\,295$ , $r = 15\text{ mm}$ )

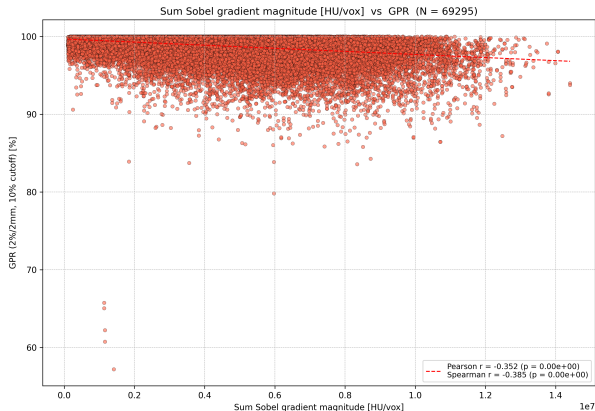


| Metric                     | Spearman      | Pearson       |
|----------------------------|---------------|---------------|
| sum_sobel_bp               | −0.385        | −0.352        |
| hu_change_per_region       | −0.339        | −0.286        |
| p95_sobel_bp               | −0.322        | −0.260        |
| $\sigma_{\text{HU}}$ at BP | −0.310        | −0.239        |
| lateral_hu_var_bp          | −0.310        | −0.267        |
| hu_change_per_mm           | −0.297        | −0.183        |
| bp_range_mm                | −0.293        | −0.047        |
| max_hu_jump                | −0.291        | −0.229        |
| max_hu_gradient            | −0.291        | −0.164        |
| total_hu_change            | −0.285        | −0.059        |
| $n_{\text{regions}}$       | −0.268        | −0.048        |
| hetero_fraction            | −0.268        | −0.143        |
| <b>pflugfelder_hi</b>      | <b>−0.242</b> | <b>−0.127</b> |
| interface_bp_dist.         | +0.141        | +0.065        |

All  $p \approx 0$ . **Bold:** new metric. Spearman  $\gg$  Pearson confirms monotonic but non-linear relationships.

**Key shift:**  $\Sigma S_{\text{BP}}$  (sum Sobel) now leads instead of  $S_{95, \text{BP}}$  — Sobel aggregation changes with sphere geometry.

# Best Predictor: Sum Sobel in BP Sphere



$\Sigma S_{BP}$  achieves Spearman  $\rho = -0.385$  (Pearson  $r = -0.352$ ,  $p \approx 0$ ,  $n = 69\,295$ ). Compare with last week:  $S_{95,BP}$  led with  $\rho = -0.391$  (BP-range,  $n = 3\,950$ ). Results are consistent across methods and at  $18\times$  larger scale.

# Motivation for Energy Stratification

**Observation** (confirmed at full scale): lower-energy bins show greater GPR variance ( $\sigma \approx 1.85\%$  at 70–100 MeV vs. 0.63% at 190–220 MeV).

**Question:** are heterogeneity–GPR correlations genuine, or are they confounded by beam energy?

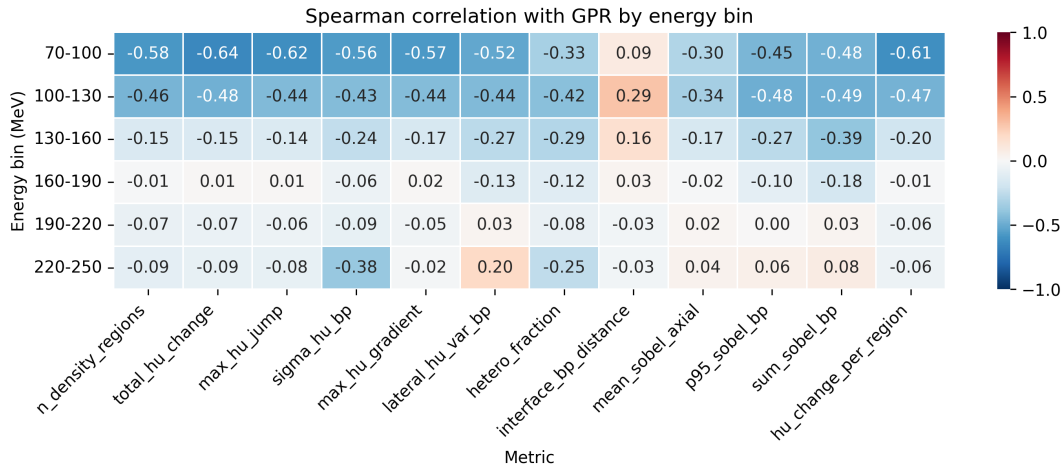
**Three complementary analyses** (same as last week, now at scale):

- 1 **Binned correlations:** Spearman  $\rho$  within each 30 MeV energy bin independently.
- 2 **Energy-coloured scatter plots:** visualise metric vs. GPR coloured by energy.
- 3 **Partial correlations:** partial Spearman controlling for energy,

$$\rho_{XY.Z} = \frac{\rho_{XY} - \rho_{XZ} \rho_{YZ}}{\sqrt{(1 - \rho_{XZ}^2)(1 - \rho_{YZ}^2)}}$$

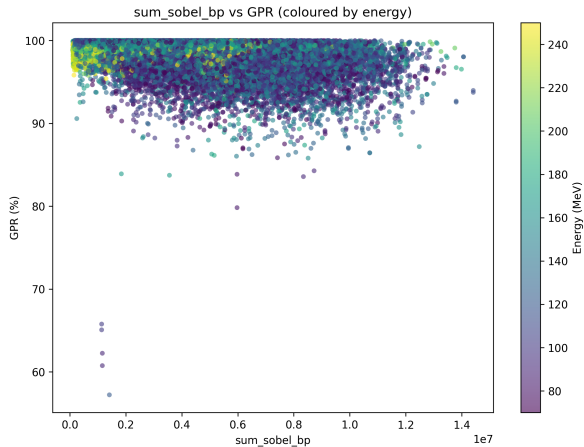
where  $X$  = metric,  $Y$  = GPR,  $Z$  = energy.

# Spearman Correlation by Energy Bin



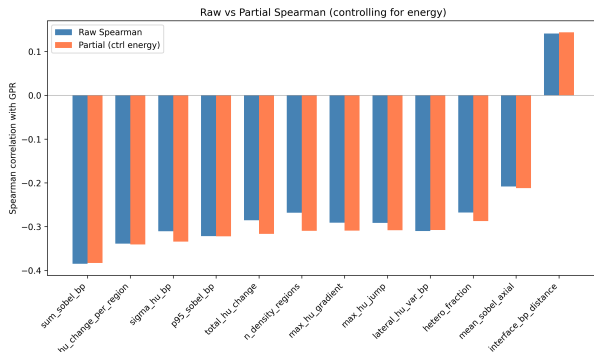
Correlations are **strongest at low energies**. 70–100 MeV:  $|\rho| > 0.60$  for HU-change metrics and  $> 0.48$  for Sobel metrics. Above 160 MeV, correlations largely vanish; at 190–220 MeV,  $\rho \approx 0$  for most metrics.

# Energy-Coloured Scatter: Sum Sobel at BP



Low-energy beamlets (dark) cluster in the high- $\Sigma S$  / low-GPR region; high-energy beamlets (bright) populate the compressed-metric / high-GPR region. The energy gradient is visible but does *not* drive the correlation (see partial analysis).

# Partial Correlations Controlling for Energy

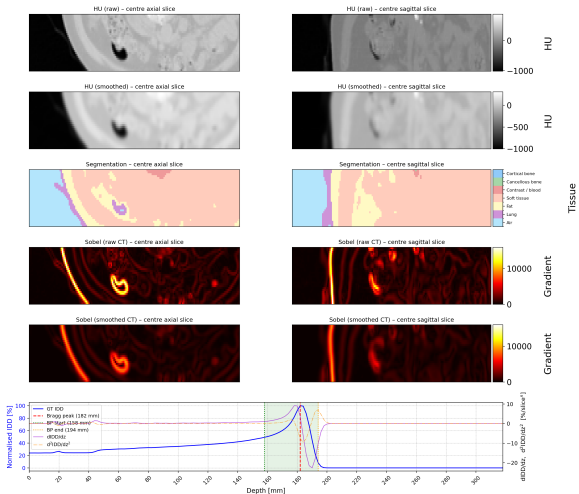


| Metric                  | Raw    | Partial |
|-------------------------|--------|---------|
| sum_sobel_bp            | −0.385 | −0.383  |
| hu_change/region        | −0.339 | −0.341  |
| $\sigma_{\text{HU,BP}}$ | −0.310 | −0.334  |
| p95_sobel_bp            | −0.322 | −0.322  |
| $n_{\text{regions}}$    | −0.268 | −0.309  |
| max_hu_gradient         | −0.291 | −0.309  |

Raw and partial correlations differ by  $|\Delta| < 0.04$ : the heterogeneity–GPR association is **not confounded by energy**. For  $n_{\text{regions}}$  ( $\Delta = -0.041$ ), controlling for energy *strengthens* the correlation, mirroring last week's finding.

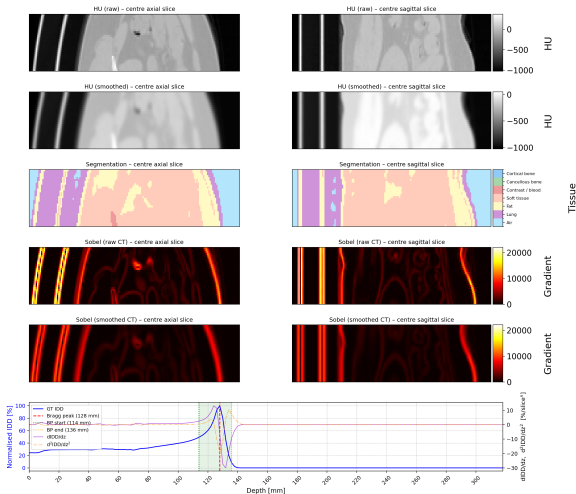
# Worst-GPR Beamlet: 1f279484, 120.67 MeV, GPR = 57.22 %

CT HU & Segmentation - 1f279484-7da5-4193-9fcb-0ac81c6bf1b6



# Best-GPR Beamlet: ffa7d740, 93.87 MeV, GPR=100.00 %

CT HU & Segmentation - ffa7d740-9119-4f00-ad47-e818e0584544



# Sphere Radius: 15 mm vs 5 mm

**Hypothesis:** a tighter sphere ( $r = 5$  mm) isolates the peak more precisely; a larger sphere ( $r = 15$  mm) captures more of the distal fall-off anatomy.

## Results pending

Run 20260421\_222950 (radius = 5 mm) is currently processing on cuda:2. Expected completion:  
**Thursday 23 April.**

This slide will be updated with:

- Side-by-side correlation table:  $r = 5$  mm vs.  $r = 15$  mm.
- Energy-stratified comparison.
- Recommendation for the default radius in future analyses.

Both radii use the same model (DoTA\_v3\_grid\_search\_v11), same dataset ( $n = 70\,145$ ), and identical metric definitions — only the spatial footprint changes.

# Key Findings

- 1 **Scale confirmed:** findings from last week ( $n = 3\,950$ ) fully replicated at  $n = 69\,295$  ( $18\times$  larger). Sobel-based metrics remain the strongest single predictors of GPR degradation.
- 2 **Sphere method:** switching from BP-range to sphere changes the leading metric from  $S_{95, \text{BP}}$  to  $\Sigma S_{\text{BP}}$  (sum vs. percentile), but both reach  $|\rho_s| \approx 0.38\text{--}0.39$  — essentially equivalent predictive power.
- 3 **Energy-stratification at scale:** correlations strongest at 70–100 MeV ( $|\rho_s| > 0.60$  for HU-change metrics), weaken above 160 MeV. Partial correlations confirm energy is *not* a confounder ( $|\Delta\rho| < 0.04$ ).
- 4 **Pflugfelder HI:** achieves Spearman  $\rho = -0.242$  (rank #13 of 17). Clinically interpretable (water-equivalent units) but weaker than Sobel metrics. WEPL std ( $\rho = -0.231$ ) is similar.
- 5 **Radius sensitivity:** pending (run2 at  $r = 5$  mm).

# Next Steps

- 1 **Sphere-radius comparison** (this week): compare  $r = 5$  mm vs.  $r = 15$  mm once run2 finishes. Determine optimal radius for downstream use.
- 2 **Composite difficulty score**: combine top metrics ( $\Sigma S_{BP}$ ,  $\sigma_{HU,BP}$ ,  $\Delta H/n_{reg}$ ) into a single “difficulty index” via PCA or logistic regression on the full  $n = 69\,295$  dataset.
- 3 **VLM difficulty estimation**: implement Track A (zero-shot) as a proof-of-concept and compare Spearman  $\rho$  against engineered metrics on a held-out subset.
- 4 **Worst-case deep-dive**: visualise and analyse the beamlets with GPR  $< 90\%$  in the 70–100 MeV bin to identify anatomical patterns driving failure.

**Questions?**



D. Pflugfelder, J. J. Wilkens, U. Oelfke, "Worst case optimization: a method to account for uncertainties in the optimization of intensity modulated proton therapy," *Phys. Med. Biol.*, vol. 53, no. 6, pp. 1689–1700, 2007.

<https://doi.org/10.1088/0031-9155/53/6/013>